Thermodynamic Properties of Pure Gases using Cubic Equations of State

Enthalpy, Entropy and Fugacity Calculations | Python Software Project | Submission date: 10 November 2026

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1. Objective

The objective of this project was to extend an equation-of-state calculation engine to evaluate fundamental residual thermodynamic properties and phase equilibrium. Developed in Python 3 using an interactive desktop graphical user interface ("Gas Property Lab" via `tkinter` and `ttk`), the application addresses two primary engineering goals:

- Determining residual enthalpy (H^R) and residual entropy (S^R) from cubic equations of state (vdW, RK, SRK, PR) using analytical departure function integrations.
- Evaluating the fugacity coefficient (φ) and fugacity (f) of pure gases to identify thermodynamically stable phases and estimate vapor-liquid saturation pressures (P^{sat}).

The software features a modular object-oriented architecture built with Python `dataclasses` (`Gas`, `State`, `Mode`) and a generalized `CubicEOS` solver class. It includes a database of 14 pure gases (Methane, Ethane, Propane, n-Butane, Ethylene, Propylene, Nitrogen, Oxygen, CO_2, CO, Hydrogen, Ammonia, Water, and Argon) across four equations of state: van der Waals (vdW), Redlich-Kwong (RK), Soave-Redlich-Kwong (SRK), and Peng-Robinson (PR).

2. Thermodynamic Theory

Following the generalized cubic formulation presented by Smith, Van Ness, and Abbott [1], all four cubic equations of state can be expressed in a unified form:

$$P = RT / (V - b) - a(T) / [(V + \varepsilon b)(V + \sigma b)] \quad (1)$$

$$b = \Omega \cdot R \cdot T_c / P_c , \qquad a(T) = \Psi \cdot \alpha(T_r) \cdot R^2 \cdot T_c^{\,2} / P_c \quad (2)$$

Table 1: Parameters and temperature dependence for the four cubic equations of state.

Equation of State	**σ**	**ε**	**Ω**	**Ψ**	**α(T_r) and κ parameter**
van der Waals (vdW)	0	0	1/8 (0.1250)	27/64 (0.42188)	$\alpha = 1$
Redlich-Kwong (RK) [4]	1	0	0.08664	0.42748	$\alpha = T_r^{-0.5}$
Soave-Redlich-Kwong (SRK) [3]	1	0	0.08664	0.42748	$\alpha = [1 + \kappa(1 - \sqrt{T_r})]^2$, $\kappa = 0.480 + 1.574\omega - 0.176\omega^2$
Peng-Robinson (PR) [2]	$1 + \sqrt{2}$	$1 - \sqrt{2}$	0.07780	0.45724	$\alpha = [1 + \kappa(1 - \sqrt{T_r})]^2$, $\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$

Introducing the dimensionless groups $\beta = bP / (RT)$ and $q = a / (bRT)$ transforms the equation of state into a cubic polynomial in compressibility factor Z:

$$Z^3 + [(\sigma + \varepsilon)\beta - (1 + \beta)]\, Z^2 + [\sigma\varepsilon\beta^2 - (1 + \beta)(\sigma + \varepsilon)\beta + q\beta]\, Z - [(1 + \beta)\sigma\varepsilon\beta^2 + q\beta^2] = 0 \quad (3)$$

When solved analytically, the largest real root represents the vapor phase, while the smallest real root that satisfies the physical limit $Z > \beta$ corresponds to the liquid phase.

Residual Departure Functions

Residual enthalpy, entropy, and fugacity coefficient are derived by integrating the volumetric departures with respect to pressure:

$$I = [1 / (\sigma - \varepsilon)] \cdot \ln[(Z + \sigma\beta) / (Z + \varepsilon\beta)] \quad \text{(for vdW: } I = \beta / Z) \tag{4}$$

$$H^R / (RT) = Z - 1 + [d \ln \alpha / d \ln T_r - 1] \cdot q \cdot I \tag{5}$$

$$S^R / R = \ln(Z - \beta) + [d \ln \alpha / d \ln T_r] \cdot q \cdot I \tag{6}$$

$$\ln \varphi = Z - 1 - \ln(Z - \beta) - q \cdot I, \quad f = \varphi \cdot P \tag{7}$$

The derivative term $d \ln \alpha / d \ln T_r$ is equal to 0 for vdW, −0.5 for RK, and $-\kappa\sqrt{T_r} / [1 + \kappa(1 - \sqrt{T_r})]$ for SRK and PR.

Ideal Gas Contribution and State Changes

Total enthalpy and entropy are determined relative to the reference state (T_0 = 298.15 K, P_0 = 1.0 bar, where $H^{ig} = S^{ig} = 0$) using temperature-dependent ideal gas heat capacity polynomials $C_p^{ig} / R = A + BT + CT^2 + D/T^2$:

$$H^{ig} = R [A(T - T_0) + (B/2)(T^2 - T_0^2) + (C/3)(T^3 - T_0^3) - D(1/T - 1/T_0)] \tag{8}$$

$$S^{ig} = R [A \ln(T/T_0) + B(T - T_0) + (C/2)(T^2 - T_0^2) - (D/2)(1/T^2 - 1/T_0^2)] - R \ln(P/P_0) \tag{9}$$

Total properties are $H = H^{ig} + H^R$ and $S = S^{ig} + S^R$. For any process between State 1 (T_1, P_1) and State 2 (T_2, P_2), the property changes are $\Delta H = \Delta H^{ig} + H^R_2 - H^R_1$ and $\Delta S = \Delta S^{ig} + S^R_2 - S^R_1$.

3. Program Structure and Workflow

The Python application executes calculations through the following systematic procedure:

1. Gas selection and operational parameters (T, P) are received from the Tkinter GUI.
2. Reduced properties ($T_r = T / T_c$, $P_r = P / P_c$) and EOS parameters (β, q, α) are evaluated.
3. Polynomial coefficients are constructed, and cubic roots for Z are solved analytically in `solve_cubic()` (using the trigonometric method when three real roots exist, and Cardano's method for a single real root).
4. Physical roots satisfying $Z > \beta + 10^{-9}$ are filtered and sorted.
5. For each root, molar volume $V = Z \cdot R_{vol} \cdot T / P$, residual properties H^R and S^R, ln φ, and fugacity $f = \varphi \cdot P$ are evaluated.
6. When both vapor and liquid roots exist, the thermodynamically stable root is identified by comparing Gibbs energy departures (the state with the lower fugacity is selected).
7. Ideal-gas enthalpy and entropy departures are computed and added to obtain total H and S.
8. For vapor-liquid saturation (`saturation()`), successive substitution $P_{new} = P \cdot (\varphi^L / \varphi^V)$ is iterated starting from the Wilson correlation until convergence $|\text{ratio} - 1| < 10^{-10}$ is achieved.

Table 2: Architecture and functional responsibilities in the Python implementation.

Component / Function	Description / Implementation Detail	Module Role
`Gas` (dataclass)	Immutable storage of critical constants (T_c, P_c), acentric factor ω, and ideal gas heat capacity polynomial parameters (a, b, c, d).	Data Model
`State` (dataclass)	Stores the complete thermodynamic result for one root: Z, V, H^R, S^R, ln φ, φ, and fugacity f.	Data Model
`solve_cubic()`	Analytically calculates the roots of $x^3 + c_2x^2 + c_1x + c_0 = 0$ using Cardano and trigonometric methods.	Math Utility

Component / Function	Description / Implementation Detail	Module Role
`CubicEOS` (class)	Encapsulates equation parameters (σ, ε, Ω, Ψ), $\alpha(T_r)$, $d \ln \alpha / d \ln T_r$, root extraction, state evaluation, phase stability determination, and VLE saturation iteration.	EOS Engine
`ideal_h()`, `ideal_s()`	Integrates C_p^{ig} polynomials relative to T_{ref} = 298.15 K and P_{ref} = 1.0 bar.	Thermodynamics
`ThermoApp` (class)	Tkinter/ttk desktop GUI featuring 5 calculation modes, Treeview result grids, KPI summary cards, input validation, and CSV export.	User Interface

The interface supports five operational modes: (1) *Properties at T, P*, (2) *Change in H and S*, (3) *Table vs pressure* (parametric sweeps with CSV export), (4) *Compare the 4 EOS*, and (5) *Saturation (Psat)*. Engineering units are K for temperature, bar for pressure, cm^3/mol for molar volume, J/mol for enthalpy, and J/(mol·K) for entropy.

4. Computational Results and Benchmark Examples

Example 1: Single State Calculation (Mode 1: Properties at T, P)

Evaluating propane using the Peng-Robinson EOS at T = 300 K and P = 5 bar. The cubic equation yields three physical roots; comparing fugacity values indicates that the vapor root is stable (f = 4.602 bar < 8.268 bar):

Table 3: Root comparison and residual properties for propane at 300 K and 5 bar (PR EOS).

Root / State	Z	V (cm^3/mol)	H^R (J/mol)	S^R (J/(mol·K))	ln φ	φ	f (bar)
Vapor (stable)	0.91445	4561.6	−587.7	−1.2695	−0.0829	0.9204	4.602
Liquid (metastable)	0.01749	87.3	−16018.9	−57.578	0.5030	1.6536	8.268

Total properties relative to reference conditions (298.15 K, 1 bar): H = 138.9 + (−587.7) = −448.8 J/mol; S = −12.92 + (−1.27) = −14.19 J/(mol·K).

Example 2: Change of State (Mode 2: Change in H and S)

Propane (PR EOS) transitioning from State 1 (300 K, 1 bar) to State 2 (400 K, 20 bar):

- Ideal-gas baseline: ΔH^{ig} = 8479.8 J/mol, ΔS^{ig} = −0.641 J/(mol·K).
- Residual corrections: H^R_1 = −110.9 J/mol, H^R_2 = −1581.2 J/mol; S^R_1 = −0.235 J/(mol·K), S^R_2 = −2.745 J/(mol·K).
- Net real gas changes: ΔH = 7009.5 J/mol, ΔS = −3.151 J/(mol·K). The real-gas departure reduces the required enthalpy change by approximately 17.3% compared to an ideal gas.

Example 3: Benchmark Comparison of the Four EOS (Mode 4)

Carbon dioxide at T = 350 K and P = 50 bar evaluated across all four models:

Table 4: Comparison of the four cubic equations of state for CO_2 at 350 K and 50 bar.

Equation of State	Phase	Z	V (cm^3/mol)	H^R (J/mol)	S^R (J/(mol·K))	φ	f (bar)
van der Waals (vdW)	Gas	0.8389	488.20	−1217.6	−2.224	0.8599	42.99
Redlich-Kwong (RK)	Gas	0.8349	485.90	−1515.0	−3.009	0.8533	42.66
Soave-Redlich-Kwong (SRK)	Gas	0.8491	494.19	−1686.8	−3.602	0.8638	43.19
Peng-Robinson (PR)	Gas	0.8292	482.57	−1758.8	−3.631	0.8456	42.28

Parametric Pressure Sweeps (Mode 3: Table vs Pressure)

Table 5: Tabulated pressure sweep results exported directly from the Python GUI application.

P (bar)	Propane (PR, 400 K) Z	Propane (PR, 400 K) H^R (J/mol)	Propane (PR, 400 K) φ	CO_2 (vdW, 350 K) S^R (J/(mol·K))	CO_2 (PR, 350 K) S^R (J/(mol·K))	CO_2 (PR, 350 K) f (bar)
1.0	0.994	−85.2	0.995	−0.05	−0.07	0.99
10.0	0.942	−870.4	0.948	−0.48	−0.72	9.51
20.0	0.875	−1780.1	0.892	−1.02	−1.55	18.25
30.0	0.798	−2710.6	0.828	−1.60	−2.51	26.24
40.0	0.680	−3720.0	0.741	−2.22	−3.63	33.48

Physical Observations: As pressure increases, molecular proximity intensifies attractive forces, decreasing Z from near unity down to 0.680 at 40 bar. In CO_2, Peng-Robinson predicts a sharper entropy departure than van der Waals (−3.63 vs −2.22 J/(mol·K)), and the fugacity deviates downward from the ideal gas line (f = P).

5. Verification and Validation

The implementation was verified using four independent thermodynamic checks:

- **Thermodynamic Consistency:** For all 14 gases across all 4 EOS, H^R calculated via the analytical departure formula was compared against numerical differentiation $-RT^2 (\partial \ln \varphi / \partial T)_P$ using central finite differences ($\Delta T = 0.01$ K). In addition, residual entropy was verified against $S^R = (H^R - G^R) / T$ where $G^R = RT \ln \varphi$. The maximum relative discrepancy observed across 224 evaluated test states was 2.5×10^{-4}, confirming mathematical consistency.
- **Ideal Gas Limiting Behavior:** In the low-pressure limit ($P \to 0$ bar), the program verified that $Z \to 1.000$, $\varphi \to 1.000$, $H^R \to 0$ J/mol, and $S^R \to 0$ J/(mol·K).
- **Phase Equilibrium Convergence:** The vapor-liquid saturation solver converged for all 392 evaluated test cases (14 gases × 4 EOS × 7 reduced temperatures ranging from $T_r = 0.50$ to 0.99), consistently satisfying the iso-fugacity criterion $|f^V - f^L| / f < 10^{-10}$.
- **Comparison with Experimental Data:** Saturation pressure and enthalpy of vaporization from the PR equation were benchmarked against literature values (NIST Chemistry WebBook / Perry's Chemical Engineers' Handbook):

Table 6: Benchmark comparison of calculated saturation parameters against experimental literature data.

Test Case	P^{sat} (EOS)	P^{sat} (Exp.)	ΔH_{vap} (EOS)	ΔH_{vap} (Exp.)	Relative Error
Propane at T = 300 K	9.99 bar	9.97 bar	14.75 kJ/mol	14.80 kJ/mol	< 0.5%
Water at T = 373.15 K	0.960 bar	1.013 bar	42.10 kJ/mol	40.70 kJ/mol	~ 5.2%

The agreement for non-polar hydrocarbons like propane is excellent (within 0.5%), while polar fluids like water exhibit expected deviations (~5%) characteristic of standard two-parameter cubic equations of state without polar association terms.

6. Discussion and Conclusions

The project results illustrate the practical significance of real gas departures in chemical process calculations. For carbon dioxide at 350 K and 50 bar, compressibility factors ($Z \approx 0.83$–0.85) and fugacity coefficients ($\varphi \approx 0.85$–0.86) reveal marked non-ideality, causing the effective thermodynamic pressure (fugacity ≈ 42–43 bar) to differ significantly from the mechanical pressure. The negative values of residual enthalpy and entropy confirm that molecular attraction predominates under these dense gaseous conditions.

Equations of state that incorporate the acentric factor (SRK and PR) produce larger and more accurate residual corrections than classical vdW and RK models because they account for molecular non-sphericity. In phase equilibrium calculations, comparing fugacities correctly isolates the thermodynamically stable root. The developed Python desktop application fulfills all project requirements: complete cubic-EOS residual property evaluation, pure component fugacity and saturation calculations, automated phase identification, an interactive Tkinter graphical interface, and CSV file reporting.

Model Limitations: The heat capacity polynomials are valid over the specified temperature ranges (typically 298–1500 K). Furthermore, classical cubic equations of state lose quantitative accuracy in the vicinity of the critical point (where density fluctuations dominate) and for strongly associating polar fluids (such as water and ammonia) or quantum gases (such as hydrogen) where advanced association or quantum corrections are required.

References

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